

Validated and Optimized Experimental Research Protocol

Combined FGF17 and 7,8-Dihydroxyflavone (7,8-DHF) Across the Rat Lifespan: a Factorial Test of Interaction on Cognition, Neuroplasticity and Oligodendrogenesis

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Document	Final protocol v5, after five rounds of multi-agent validation
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Validation method	Per round: 3 independent Claude Opus critics (study design and biostatistics; pharmacology and delivery; behaviour, myelin and plasticity) plus 1 Claude Haiku reference librarian. 5 rounds (20 agent reviews) plus one dedicated Opus review of the lifespan module.
Status	Not an approved protocol. No reviewer gave an unconditional APPROVE in any round. Final status: confirmatory core "approve with changes", conditional on pilot gates; lifespan module exploratory only.

Executive summary

The source document claimed to be "fully validated" after 15 rounds. When it was audited, it contained errors that made the experiment impossible or uninterpretable as written: the stated 240 rats were 120 by its own group list; the interaction test was powered at 0.60, not 0.80; the Alzet 1007D pump cannot run 14 days; the behavioural battery was stacked onto one day; the "factorial" contained orphan arms; 17% DMSO daily for 28 days threatened the myelin endpoint it was meant to measure; and three of the seven references (Wang 2026, Gardner 2026, Andero/Choi/Ressler 2014) could not be found.

The protocol was rebuilt over five rounds. The final design is:

- **Confirmatory core (Exp 1):** a 2x2 factorial (FGF17 × 7,8-DHF) in **aged (≥22 months) rats**, 32 completers per cell (216 enrolled at 40% assumed attrition), one primary endpoint (Morris water maze probe, day 26), one pre-specified interaction contrast, and matched exposure windows (28-day ICV pump and 28 days of 7,8-DHF).
- **Mechanism and engagement:** satellite cohort for TrkB signalling, aged-slice LTP cohort, an ANA-12 TrkB-necessity experiment (hypothesis-generating), and an off-drug retention probe.
- **Lifespan module (Exp 5, added at the researcher's request):** neuroplasticity and cognition across age from newborn to oldest rats, re-scoped after critic review into (a) infant/juvenile plasticity and myelination arms (P10, P21; no cognition tests, because hippocampus-dependent tasks are not valid at those ages) and (b) a four-age cognition series by age at treatment onset (P28, 6, 12 and ≥22 months). Declared exploratory and descriptive.
- **Stage-gating:** eight pilots (P1-P8) with explicit go/no-go rules must pass before the confirmatory study enrolls. Core = 488 animals; full programme = 992 animals.

Bottom line on remaining gaps (section 9): yes, items remain that the critics do not like. None is an arithmetic or logic error any more, but three are blocking until pilots are run (FGF17 28-day bioactivity, cannula fixation in growing rats, neonatal ICV safety), the 7,8-DHF mechanism is still disputed, and the primary interaction test is powered only for a medium-large effect ($f \geq 0.25$).

1. Validation history: five rounds

Each round: three Opus critics reviewed the current version independently (they were told to be hostile and to recompute all numbers); a Haiku agent searched for and checked the literature. All reference checks were based on web-search snippets because PubMed, PMC, Crossref and publisher hosts were unreachable from the sandbox; see section 10 for what that means.

Round	Version reviewed	Verdicts	Most important findings fixed in the next version
1	v0 (source doc)	Design: REJECT. Pharmacology: REJECT. Behaviour: REJECT.	Total N 240 was really 120; interaction power 0.60 (not 0.80); aged arm had no control cells and a different route (intranasal vs ICV); Alzet 1007D lasts ~7 d, not 14; 17% DMSO daily × 28 d threatens myelin; R13 dose not molar-matched; battery impossible (all tasks on day 35; "remote" memory with 3-day delay); no randomisation, blinding or primary endpoint; no rat FGF17 ELISA exists; BrdU/EdU schedule missing; no causal (necessity) test; 3 of 7 references not found.
2	v1	Design: APPROVE WITH CHANGES. Pharmacology: APPROVE WITH CHANGES. Behaviour: APPROVE WITH CHANGES.	FGF17 off-board at the day-26 primary endpoint while DHF on-board; PD173074 ICV arm infeasible and logically circular (it would block FGF17 itself); EdU 50 mg/kg × 7 is cytotoxic; DHF vehicle unverified; confirmatory population (young rats) was the weakest-supported prediction; necessity arms lacked antagonist-alone controls; IHC and EM cannot share one perfusion; 18–20-month Wistar rats are middle-aged.
3	v2	All three: APPROVE WITH CHANGES.	42 enrolled × (1–0.30) = 29, not 32 completers; 30% attrition unrealistic and informative (MNAR); odd-numbered tissue cells cannot be sex-balanced; reversal/CFC fell after the drug stopped; Exp 4 run in young rats cannot test an aged effect; FGF17 28-day stability needs pass/fail rules; ANA-12 gate and EdU labelling adequacy; aged water-maze hypothermia controls.
4	v3 (+ lifespan draft)	All three: APPROVE WITH CHANGES. Lifespan-module critic: REJECT as drafted.	Mid-study pump exchange in aged rats; no off-drug washout probe; aged NOR may floor and block gatekeeping; ANA-12 gate measured without DHF on board; EdU gate defined in young not aged tissue; Exp 4 power hidden (0.50 at f=0.25); equivalence margin coincided with detectability limit. Lifespan draft: dosing window and test age confounded; cognition tasks invalid before ~P21; unsafe ICV route harmonisation in pups.
5	v4	Pharmacology: APPROVE WITH CHANGES. Design: conditional APPROVE of confirmatory core. Behaviour: APPROVE WITH MINOR REVISIONS (core); lifespan module exploratory only.	Applied in this document (v5): undefined LTP cohort; fixed FGF17 dose in growing P28 rats; tier-2 stability fallback; cannula fixation over skull growth; non-uniform attrition assumptions; gatekeeping order; litter clustering; DHF/FGF17 exposure alignment; honest H1b wording; heparin-free potency; rat DHF dose literature.

No round produced an unconditional APPROVE. Rounds 2–5 progressively moved from REJECT to "approve with changes"; the changes requested in each round were made before the next, and numbers were independently recomputed by the critics in rounds 2–5 (all totals and powers in this document were reproduced).

2. Hypotheses and endpoints

- **H1 (confirmatory, aged ≥ 22 months):** positive FGF17 $\times$ 7,8-DHF interaction on Morris water maze (MWM) probe-trial percentage of time in the target quadrant (day 26, on treatment). "Synergy" is claimed only if the interaction confidence interval excludes zero and the combination exceeds each single agent.
- **H1b (exploratory, tested in parallel, outside the gatekept chain):** the same contrast at a post-washout retention probe (day 31; ≥ 72 h after the last 7,8-DHF dose; FGF17 pump expired on day 28, so residual peptide cannot be excluded). Wording is "post-washout retention", not "durable structural effect"; the day-26 probe is itself an unreinforced extinction trial and is entered as a covariate.
- **H2 (exploratory):** young adults (6 months) show a smaller effect (ceiling).
- **H3 (hypothesis-generating):** FGF17 increases new oligodendrocytes (EdU+/CC1+); ANA-12 attenuates the aged combination effect (acute signalling dependence only).
- **H4a (exploratory, within-age):** in infant (P10) and juvenile (P21) rats, how FGF17, 7,8-DHF and the combination change plasticity (LTP, spines, TrkB signalling, parvalbumin/perineuronal nets) and myelination.
- **H4b (exploratory, descriptive):** how the effect on cognition and structure varies with age at treatment onset (P28, 6, 12, ≥ 22 months). H4a and H4b are separate questions; they use different routes and durations and are never pooled.

Fixed-sequence gatekept key secondaries (each tested only if the previous passes, $\alpha = .05$): MWM acquisition path length (mixed model) $\rightarrow$ g-ratio (EM) $\rightarrow$ CA1 spine density $\rightarrow$ EdU+/CC1+ density $\rightarrow$ Y-maze alternation $\rightarrow$ PSD-95/synaptophysin $\rightarrow$ NOVEL OBJECT RECOGNITION (only if aged-NOR pilot P5 passes). Contextual fear conditioning and H1b are exploratory and sit outside the chain. qPCR (Mbp, Olig1/2, Pdgfra, Srf; ≥ 3 validated reference genes, MIQE) uses Benjamini-Hochberg FDR. Structural endpoints are placed ahead of fragile ones so that a floored behavioural or low-yield EdU readout does not forfeit robust endpoints.

3. Design, animals and power

Wistar rats of both sexes (1:1, stratified), computer-randomised (stratified by sex, litter, cage), allocation concealed, coded vials, automated tracking, blinded histology. Pre-specified exclusions: pump/cannula misplacement (dye + atlas check, with tip verification at termination), $>15\%$ weight loss, MWM swim speed $<50\%$ of cohort median, visible-platform failure, vision/motor prescreen failure, hypothermia rule, humane endpoints.

Experiment	Population / design	Enrolled	Completers (assumed attrition)	Status
Exp 1 – CONFIRMATORY	Aged ≥ 22 mo; 2 $\times$ 2 FGF17 $\times$ 7,8-DHF; 4 cells	216 (54/cell)	128 (32/cell; 40%)	Confirmatory
Exp 1S – satellite	Aged; same dosing, no behaviour; pTrkB/pERK/pAKT/pPLC γ at 1 h (peak) and 18 h (trough)	80 (20/cell)	48 (12/cell, 6 per time point, 3/sex)	Mechanistic gate
Exp 1L – LTP cohort	Aged; same dosing, no behaviour; ex vivo CA1 LTP (own animals, terminal)	56 (14/cell)	32 (8/cell)	Gated by pilot P8
Exp 4 – ANA-12	Aged; Veh+saline, Veh+ANA-12, Combo+saline, Combo+ANA-12	108 (27/arm)	64 (16/arm)	Hypothesis-generating
Exp 2 – young adult	6 mo; 2 $\times$ 2	80 (20/cell)	64 (16/cell; 20%)	Exploratory
Exp 3 – R13 reference	6 mo; R13 14.5 mg/kg PO vs gavage vehicle	32 (16/arm)	32	Exploratory
Exp 5A – infant/juvenile plasticity	P10 and P21; 2 $\times$ 2; split-litter; no cognition	80 pups (40/age)	64	Exploratory
Exp 5B – P28-onset cohort	Onset at P28 (tested $\sim$ P43–P61); 2 $\times$ 2; same pump + battery as Exp 1	80 (20/cell)	64 (16/cell; 20%)	Exploratory
Exp 5C – 12-month cohort	12 mo; 2 $\times$ 2; same pump + battery	80 (20/cell)	64 (16/cell; 20%)	Exploratory
Experimental total		812		
Pilots P1–P8	See section 5	180		Mandatory gates
GRAND TOTAL		992		

Minimum viable confirmatory core (Exp 1 + 1S + 1L + pilots P1–P5 and P8) = 216 + 80 + 56 + 130 + 6 = **488 animals**. Exp 2–5 are stage-gated: they enrol only after the pilots that guard Exp 1 (P2, P3b, P4, P5, P8) have passed. A multi-year, multi-site programme is realistic for 992 animals; the staged plan is mandatory, not

optional.

Power (completers, 2×2 interaction contrast, 1 df, noncentral F).

Test	N	f = 0.15	f = 0.20	f = 0.25	f = 0.30	f = 0.40	f = 0.50
Exp 1 primary, $\alpha = .05$	128	0.39	0.61	0.80	0.92	–	–
Exp 1 primary, $\alpha = .05$, fewer completers	116	–	–	0.76	–	–	–
Exp 1 primary, $\alpha = .01$	128	–	–	0.59	–	–	–
Exp 4 and per-age cells (Exp 2, 5B, 5C)	64	–	–	0.50	0.66	0.88	0.98
Exp 3 (two-group t test, n = 16/arm)	32	d = 0.8: 0.59	d = 1.0: 0.78	d = 1.2: 0.91	–	–	–

The primary test is powered only for $f \geq 0.25$ ($\approx |d| 0.5$ on the 1-df contrast). The anchor is the main-effect size reported in aged mice (Iram 2022, $d \approx 1.2$), which is not an interaction effect size and was not measured in rats; interaction effects are usually smaller than main effects, so a real synergy may be undetectable. Null results are reported with intervals and described as inconclusive; no equivalence claim is made. Attrition: 40% for aged rats (vendor Wistar Han 104-week survival is 72% in males and 65% in females, plus surgical and 28-day dosing losses); 20% for young cohorts (to be confirmed by pilot experience). Primary analysis is on completers, with a tipping-point/pattern-mixture sensitivity analysis for informative dropout.

4. Interventions, timeline, endpoints, statistics

4.1 Interventions

Agent	Specification (final)
FGF17	Recombinant mouse FGF17 (R&D; 7400-FG-025/CF) in aCSF + 0.1% BSA + 10 µg/ml heparin (identical vehicle in all arms); Alzet 2004 pump (0.25 µl/h, ~28 d), right lateral ventricle, coordinates scaled to bregma-lambda distance with dye/atlas verification; 50 µg/ml ≈ 0.3 µg/day (final concentration set by dose-finding pilot P2). Mouse/rat protein identity to be confirmed by sequence alignment attached before registration and by rat-cell bioassay in P2. Meloxicam and local anaesthetic; anti-FGF17 serum antibodies at end.
FGF17 fallback (pre-specified)	Tier 1 (if P2 shows <70% bioactivity at day 28): two back-to-back Alzet 2002 pumps (25 µg/ml at 0.5 µl/h), exchange at day 14 for ALL animals, battery start moved to day 21. Tier 2 (if day-14 stability also fails): stabilised/thiol-free or heparin-optimised FGF17 variant, bioassayed at the same time points. Tier 3: repeated ICV bolus via chronic guide cannula (pharmacokinetics validated first). Any tier change re-registers the protocol before Exp 1 enrolls; there is no mid-study exchange applied to a subset.
7,8-DHF	Intraperitoneal, daily days 0–28 after each day's session (fixed afternoon time). Pilot P1 tests 10 and 5 mg/kg (published rat IP doses include 5 and 20 mg/kg) and two vehicles (≤5% DMSO + 20% HP-β-cyclodextrin; fallback 10% DMSO / 40% PEG300 / 5% Tween-80, always vehicle-matched); sciatic-nerve and body-weight safety monitoring. Half-life in mice is ~2 h, so once-daily dosing gives transient engagement; the 1 h and 18 h satellite samples are framed as peak and trough.
R13 (prodrug)	14.5 mg/kg orally daily for 28 d (molar match to 10 mg/kg 7,8-DHF: MW 368.35 / 254.24 = 1.449). Chen 2018 used 7.25, 21.8 and 43.6 mg/kg/day for 3 months in 5XFAD mice, so 14.5 mg/kg lies inside the tested range. Gavage-vehicle control arm.
ANA-12 (Exp 4)	0.5 mg/kg IP (mouse-derived) 30 min before MWM acquisition and probe sessions only; dose/schedule must give ≥50% pTrkB/pERK inhibition spanning the session <u>on a 7,8-DHF background</u> (pilot P4, with session-end time point), otherwise repeat dosing. Scope is acute-signalling dependence only.
EdU	Dose chosen in young saturation/toxicity pilot P3a (25 vs 50 mg/kg, scored against Ki67 and γH2AX); pulses on days 2, 4, 6; 27-day chase. Detectability in aged vehicle brain pre-tested in P3b (numeric rule in section 5).

4.2 Timeline for Exp 1 (day 0 = pump implant and first dose)

Days	Activity
0–13	Dosing, handling. EdU pulses on days 2, 4, 6.
14	Prescreen: vision, motor, baseline core temperature.
15 / 16	Y-maze / open field.
17–19	NOR habituation (17), sample + 5 min test (18), 24 h test (19) — analysed in the chain only if pilot P5 passed.
20	Visible-platform control.
21–25	MWM acquisition, 4 trials/day; water 25 ± 1 °C; core temperature before/after; warm rest cage (32 °C); inter-trial interval ≥5 min; floating, thigmotaxis and swim speed recorded; post-session temperature <34.5 °C triggers rest-day/exclusion rule.
26	MWM probe (PRIMARY, on treatment).
27	Rest day.
28	Contextual fear conditioning training (exploratory); last 7,8-DHF dose after the session; pump expires.
29	Fear-conditioning recall (24 h). No dosing.
31	Post-washout retention probe (H1b, exploratory).
33	Euthanasia and tissue collection (≥48 h after last behaviour).

The two agents are now time-aligned: both are delivered on days 0–28; recall (day 29) and the retention probe (day 31) are post-exposure. Order of tasks is least to most stressful, with a rest day before fear conditioning to limit swim-stress carry-over.

4.3 Tissue allocation and endpoints (per Exp 1 cell, 32 completers; every number even, sex-balanced)

Modality	n/cell	Readouts
Perfusion-fixed, PFA (IHC)	12	EdU click chemistry + CC1/Olig2/NG2 (new oligodendrocytes and OPCs), MBP, PSD-95/synaptophysin; optical-fractionator stereology; unstained autofluorescence (lipofuscin) controls
Perfusion-fixed, glutaraldehyde (EM)	8	g-ratio versus axon diameter regression, corpus callosum and CA1 stratum radiatum, ≥150 axons/animal, systematic-random sampling, blinded
Fresh, Golgi-Cox	6	CA1 spine density and subtype (Imaris)
Fresh-frozen, qPCR	6	Mbp, Olig1/2, Pdgfra, Srf; ≥3 reference genes (geNorm/NormFinder), MIQE

Each animal contributes to one modality (histology fixation methods are incompatible). Histology is a mechanistic secondary and is not powered for the interaction. Satellite engagement gate: pTrkB (Y816; antibody

validated with BDNF-stimulated positive control and peptide block), pERK, pAKT and pPLC γ at 1 h and 18 h; TrkB-specific mechanistic claims are withdrawn unless ≥ 2 orthogonal readouts agree ("hard stop"), and any 7,8-DHF effect is then reported as an effect of unestablished mechanism. EM minimum detectable difference in g-ratio is computed from pilot between-animal variance and reported as descriptive.

4.4 Statistical analysis plan (pre-register on OSF before data collection)

- **Primary:** two-factor ANOVA of probe time, with sex as a blocking factor and one pre-specified FGF17 $\times$ 7,8-DHF interaction contrast at $\alpha = .05$ (no multiplicity because it is the single confirmatory test).
- **Acquisition:** linear mixed model (day $\times$ FGF17 $\times$ 7,8-DHF, random subject) with swim speed, thigmotaxis and core temperature as covariates.
- **Secondaries:** fixed-sequence gatekeeping (section 2); molecular endpoints with Benjamini-Hochberg FDR; sex $\times$ treatment interactions exploratory.
- **Missing data:** primary on completers; tipping-point/pattern-mixture sensitivity analysis for informative dropout (the earlier "missing at random" claim was removed).
- **Exp 2-5:** standardised effect sizes with 95% intervals; litter (or litter of origin) as a random effect in 5A, 5B, 5C and Exp 2 (no more than two siblings per cell); no per-age synergy claims; no inferential age $\times$ treatment test across non-concurrent cohorts (cohort/batch is confounded with age and is recorded).
- **Software:** R (lme4, afex, emmeans, mgcv for descriptive age curves).

5. Pilot programme and go/no-go gates (must pass before Exp 1 enrolls)

Pilot	Question and design	N	Go/no-go rule
P1	DHF vehicle solubility/tolerability at 10 and 5 mg/kg; two vehicles (young)	12	Stable solution at injection concentration; no weight loss >10%, no nerve-histology change; otherwise lower dose/alternate vehicle (R13 re-matched).
P2	FGF17 dose-finding in aged rats (vehicle, 0.1×, 1×, 3×; 10 enrolled each) + in vitro 37 °C pump-mimic stability at day 0/7/14/21/28 + rat-cell bioassay + heparin-free formulation bioassay	40	≥70% bioactivity at day 28 (else tier fallback, section 4.1); in vivo pERK/EdU response selects concentration; mouse FGF17 shows rat-cell potency.
P3a	EdU saturation/toxicity in young rats (vehicle, 25, 50 mg/kg) against Ki67 and γH2AX	24	Dose that labels near-saturation without γH2AX/survival signal.
P3b	EdU detection in AGED vehicle brain (two doses)	12	Mean EdU+/CC1+ density must give ≥80% power to detect a 50% increase at the pilot SD with IHC n ≤ 12 per cell; otherwise raise IHC n or demote EdU to exploratory. Unstained autofluorescence controls mandatory.
P4	ANA-12 inhibition of pTrkB/pERK in aged rats with and without 7,8-DHF on board; time points 30 min, 2 h, 6 h and session end	32	≥50% inhibition spanning a full session on a DHF background; otherwise repeat dosing or drop the TrkB-necessity claim.
P5	Aged-rat NOR validation (longer sample, shortened interval)	10	Vehicle discrimination index >0.1 and total exploration ≥20 s; otherwise NOR is exploratory.
P6	Pump/cannula feasibility and fixation in P28 rats (28 d growth); tip accuracy by histology	12	≥80% correct ventricular tips at day 28 and acceptable welfare; otherwise Exp 5B is dropped or redesigned.
P7	Neonatal/juvenile vehicle and route tolerability; establish ICV technique, mortality and weight; heparin-free FGF17 tolerance	32	Mortality ≤15% and weight deficit ≤10%. If the single ICV bolus cannot be made safe, Exp 5A is run with 7,8-DHF only (FGF17 omitted) and declared as such.
P8	Aged slice viability and LTP magnitude (ex vivo CA1)	6	Reproducible LTP in aged vehicle slices; otherwise Exp 1L is deleted.

6. Exp 5 - Lifespan neuroplasticity and cognition (exploratory)

Question. How do neuroplasticity, myelination and cognition respond to FGF17, 7,8-DHF and their combination depending on the age of the rat, from newborn to oldest?

What the critics forced. The first draft dosed all ages for 14 days and tested at the end. The critic showed that this confounds age at dosing, age at testing, brain growth and weaning (P21), that fear conditioning (emerging ~P17–24) and the water maze (~P20–21) are not valid in infants, and that repeated ICV boluses in pups are unsafe and not comparable to a pump. The final module therefore changes the logic:

- **Cognition ages (5B, Exp 2, 5C, Exp 1):** age is defined as age at treatment onset (P28, 6, 12 and ≥22 months), with an **identical** 28-day pump/IP schedule, route and battery, so kinetics and route are not confounded with age. Limitations that remain: the P28 cohort is tested during adolescence (~P43–P61), so onset age and measurement age co-vary; body mass triples in the P28 cohort, so FGF17 exposure per kilogram and CSF concentration fall across the infusion (declared, modelled, not corrected); cohorts are not concurrent, so cross-cohort comparison is restricted to standardised effect sizes against each cohort's own vehicle arm.
- **Infant/juvenile arms (5A):** P10 (dosing window P3–P10) and P21 (P14–P21). No cognitive testing. Route is age-specific and declared as such: daily SC/IP 7,8-DHF and a single weight- and CSF-scaled ICV FGF17 bolus (heparin-free formulation tested in P2/P7). Results are analysed within age and never pooled with the pump cohorts. Split-litter design (each litter contributes one pup per cell; eight litters per age; litters standardised to 8–10 pups; maternal separation <15 min per procedure and recorded as a covariate).
- **Plasticity and structure readouts that are valid across ages:** ex vivo CA1 LTP (theta-burst) and paired-pulse (aged LTP gated by P8), Golgi–Cox spines, MBP/g-ratio and OPC/oligodendrocyte counts (rat myelination timing is region-dependent, with cortex and hippocampus starting ~P14), parvalbumin and perineuronal nets (WFA), TrkB full-length versus truncated (T1) isoform ratio, pTrkB/pERK. The statement that TrkB.T1 "rises with age" is weakly supported (a higher truncated share in adults than juveniles is reported, but no normal-ageing-into-old-age dataset exists), so the isoform work is hypothesis-generating.

- **Analysis:** treatment × age-at-onset described with standardised effect sizes and 95% intervals (four-level and log-age trend, GAM for description only); litter-level N reported for 5A; Exp 5 is not powered for three-way interactions (0.50 power for $f = 0.25$ at $n = 16$ per cell).
- **Welfare:** neonatal-specific ethics approval, hypothermic anaesthesia, maternal separation limits, pup weight tracking, humane endpoints (mortality >15% stops a cohort).

7. Animal welfare, ethics and reporting

Ethics-committee approval number, funding and conflict-of-interest statements are placeholders to be completed by the researcher before registration (neonatal procedures need their own approval). Housing 12:12 light/dark with enrichment; aged males single-housed if aggressive; weight and condition scoring; meloxicam 48 h post-surgery; estrous cytology only in cycling young females (aged females: cycle status documented). ARRIVE 2.0 Essential 10 checklist to accompany the registration (Percie du Sert 2020).

8. What happened to the claims in the source document

Original claim	Outcome
"15 rounds of agentic criticism"	Unverifiable; replaced by the documented five rounds above.
N = 240 (20 × 6 groups × 2 ages)	Wrong (120 by its own group list); replaced by explicit design and counts (section 3).
80% power for the interaction at N = 20/group	Wrong (0.60 for a 2×2 interaction at $n = 20/\text{cell}$); now 32 completers/cell for 0.80 at $f = 0.25$.
Alzet 1007D, 0.5 µl/h × 14 days	Impossible (capacity ~100 µl ≈ 7 days); now Alzet 2004 (28 days).
Intranasal FGF17, Wang 2026 (Neurotherapeutics)	Not found in any search; removed. Intranasal FGF17 is future work.
Gardner 2026 (Behav Brain Res); Andero, Choi, Ressler 2014	Not found as cited (only a 2024 preprint; a different 2012 paper); removed.
21.8 mg/kg R13 "equivalent" to 15 mg/kg 7,8-DHF	True molar equivalence, but the treatment arms used 10 mg/kg; R13 now 14.5 mg/kg. 21.8 mg/kg is one of Chen 2018's three chronic doses.
17% DMSO IP daily × 28 d	Toxic and a threat to the myelin endpoint; replaced by ≤5% DMSO + cyclodextrin (pilot-validated).
7,8-DHF is an established TrkB agonist	Disputed (Boltaev 2017; Todd 2014); now treated as a hypothesis with an engagement gate.
Budget US\$180–220k; 80% reagent saving; monkey toxicity at 30 mg/kg for 7 months	No source; removed. 992 animals make the earlier budget meaningless.

9. Remaining gaps after five rounds

Answer to the question "is there still an item or gap that the agents do not like?" Yes. After the fifth round the three Opus critics still listed the following. None is an arithmetic or logic error; the open items fall into (A) problems that only pilots can settle, (B) limits that cannot be removed by design, and (C) wording and framing changes already applied in this version.

#	Remaining gap / objection	Class	Fix
1	FGF17 may lose most activity within days at 37 °C; no 28-day FGF17 stability data exist; even the 14-day fallback is unproven.	A - blocking	Pilot P2 with pre-specified three-tier fallback. If tier 3 is needed the FGF17 limb changes substantially.
2	Cannula and pump fixation and tip accuracy in P28 rats over 28 days of skull growth.	A - blocking for Exp 5B	Pilot P6 (n = 12) with histological pass/fail.
3	No primary source found for free-hand ICV injection in P7–P10 rat pups or its mortality; no juvenile Alzet protocol found.	A - blocking for Exp 5A FGF17	Pilot P7; fallback is 7,8-DHF-only infant/juvenile arm.
4	Aged-brain EdU+/CC1+ yield may be near the floor (aged OPC cell cycle ~70 days at P240–P540); EdU measures net accumulation, not proliferation per se.	A - gated	Pilot P3b numeric rule; demote EdU endpoint if the rule fails.
5	Aged NOR may floor; aged-slice LTP has viability limits.	A - gated	Pilots P5 and P8; endpoints exploratory or deleted if they fail.
6	ANA-12 rat dose and degree/duration of TrkB blockade unknown; acute blockade may miss the chronic structural route.	A - gated, B	Pilot P4; Exp 4 stays hypothesis-generating.
7	7,8-DHF TrkB agonism is disputed; a null DHF result may reflect the compound, not the biology.	B - cannot remove	Engagement gate (satellite + P4); claims withdrawn if the gate fails.
8	Primary interaction is powered only for $f \geq 0.25$; realistic synergy may be smaller (power 0.61 at $f = 0.20$, 0.39 at $f = 0.15$); detecting $f = 0.20$ needs 50 completers/cell (200 total).	B - cannot remove without ~60% more animals	Disclosed; results framed as inconclusive when null. Increase N only if a pilot supports a smaller effect size.
9	Cognition-by-age series is non-concurrent (batch/cohort confound); the P28 cohort is tested in adolescence (onset age and measurement age co-vary); FGF17 exposure per kg falls as P28 rats grow.	B - intrinsic	Standardised effect sizes against own vehicle only; no inferential age $\times$ treatment test.
10	No FGFR-necessity experiment (PD173074 ICV infeasible and circular).	B - deferred	Follow-up with AAV-Fgfr3 shRNA or conditional knockout.
11	Mouse FGF17 in rat: identity claimed but not independently confirmed; no functional potency check yet.	C / A	Alignment attached before registration; rat-cell bioassay in P2.
12	Scale: 992 animals (core 488), ~18-month ageing pipeline for ≥ 22 -month animals, cost, ethics burden; the earlier budget cannot be used.	B	Mandatory stage-gating; Exp 2–5 only after core pilots pass.
13	Literature verification is incomplete: no full text could be opened (PubMed, PMC, Crossref and publisher hosts were unreachable); several rat-specific facts have no primary source (neonatal ICV, juvenile pumps, age-specific CSF volume, rat hippocampal TrkB isoforms, a single source for the P10–P30 myelination peak).	B	Researcher to check every reference against the publisher record before registration.

Items the critics explicitly considered **resolved** by round 5: all animal counts and power values (independently recomputed and reproduced), the timeline (no double-booked days; every primary cognitive endpoint on treatment), exposure alignment, attrition arithmetic, sex-balanced allocation, DMSO/vehicle plan, R13 dose logic, aged water-maze welfare rules, route harmonisation in Exp 1, and the removal of fabricated or unverifiable claims.

10. References and verification status

Verification was performed by a Haiku agent through web-search snippets only (WebFetch could not reach PubMed, PMC, Crossref, doi.org or most publisher sites in the sandbox). "Confirmed" therefore means consistent across several indexed records, not checked against the publisher's page. DOIs are as returned by the searches.

#	Citation	Status
1	Iram T, Kern F, Kaur A, et al. Young CSF restores oligodendrogenesis and memory in aged mice via Fgf17. <i>Nature</i> 2022;605:509–515. doi:10.1038/s41586-022-04722-0	Confirmed; Methods (dose, pump, coordinates) not accessed
2	Fortin D, Rom E, Sun H, Yayon A, Bansal R. Distinct FGF/FGF receptor signaling pairs initiate diverse cellular responses in the oligodendrocyte lineage. <i>J Neurosci</i> 2005;25:7470–7479. doi:10.1523/JNEUROSCI.2120-05.2005	Confirmed
3	Oh LY, Denninger A, Colvin JS, et al. FGFR3 signaling regulates the onset of oligodendrocyte terminal differentiation. <i>J Neurosci</i> 2003;23:883–894. doi:10.1523/JNEUROSCI.23-03-00883.2003	Confirmed
4	Chen C, Wang Z, Zhang Z, Liu X, Kang SS, Zhang Y, Ye K. The prodrug of 7,8-dihydroxyflavone development and therapeutic efficacy for treating Alzheimer's disease. <i>PNAS</i> 2018;115:578–583. doi:10.1073/pnas.1718683115 (first author is Chun Chen; R13 chronic doses 7.25/21.8/43.6 mg/kg/day × 3 mo)	Confirmed bibliographically; doses from secondary snippets
5	Liu X, et al. Biochemical and biophysical investigation of the BDNF mimetic 7,8-dihydroxyflavone in the binding and activation of the TrkB receptor. <i>J Biol Chem</i> 2014;289:27571–27584. doi:10.1074/jbc.M114.562561	Partial (author list not retrieved)
6	Boltaev U, Meyer Y, Tolibzoda F, et al. Multiplex quantitative assays indicate a need for reevaluating reported small-molecule TrkB agonists. <i>Sci Signal</i> 2017;10(493). doi:10.1126/scisignal.aal1670	Confirmed (DOI from journal URL)
7	Todd D, Gowers I, Dowler SJ, et al. A monoclonal antibody TrkB receptor agonist as a potential therapeutic for Huntington's disease. <i>PLoS One</i> 2014;9(2):e87923. doi:10.1371/journal.pone.0087923	Citation confirmed; 7,8-DHF content known only from secondary reviews
8	Do small molecules activate the TrkB receptor in the same manner as BDNF? <i>Pharmaceuticals</i> 2021;14(8):704 (review of the TrkB-agonist controversy)	Confirmed (review)
9	Lim SY, Scarlett CO, Yapici S, Ferrazzano P, Cengiz P. Pharmacokinetics of 7,8-dihydroxyflavone in neonatal mice with hypoxia-ischemia related brain injury. <i>Front Pharmacol</i> 2025;15:1508696. doi:10.3389/fphar.2024.1508696	Confirmed (mouse; single P9 5 mg/kg IP dose)
10	Cazorla M, Prémont J, Mann A, Girard N, Kellendonk C, Rognan D. Identification of a low-molecular weight TrkB antagonist with anxiolytic and antidepressant activity in mice. <i>J Clin Invest</i> 2011;121:1846–1857. doi:10.1172/JCI43992	Confirmed (mouse; no rat study found)
11	Percie du Sert N, et al. The ARRIVE guidelines 2.0. <i>PLoS Biol</i> 2020;18:e3000410. doi:10.1371/journal.pbio.3000410	Confirmed
12	Young KM, Psachoulia K, Tripathi RB, et al. Oligodendrocyte dynamics in the healthy adult CNS: evidence for myelin remodeling. <i>Neuron</i> 2013;77:873–885. doi:10.1016/j.neuron.2013.01.006	Confirmed
13	Psachoulia K, et al. Cell cycle dynamics of NG2 cells in the postnatal and ageing brain. <i>Neuron Glia Biol</i> 2009;5:57–67.	Confirmed by snippet
14	Sengupta P. The laboratory rat: relating its age with human's. <i>Int J Prev Med</i> 2013;4:624–630.	Confirmed
15	Burman MA, Erickson KJ, Deal AL, Jacobson RR. Contextual and auditory fear conditioning continue to emerge during the periweaning period in rats. <i>PLoS One</i> 2014;9:e100807. doi:10.1371/journal.pone.0100807	Confirmed
16	Rudy JW, Stadler-Morris S, Albert P. Ontogeny of spatial navigation behaviors in the rat. <i>Behav Neurosci</i> 1987;101:62–73.	Partial (abstract not seen)
17	Callahan LS, Thibert KA, Wobken JD, Georgieff MK. Early life iron deficiency anemia alters the development and long-term expression of parvalbumin and perineuronal nets in the rat hippocampus. <i>Dev Neurosci</i> 2013;35:427–436. doi:10.1159/000354178	Confirmed (iron-deficiency model, not normal development)
18	Shankar S, Teyler TJ, Robbins N. Aging differentially alters forms of long-term potentiation in rat hippocampal area CA1. <i>J Neurophysiol</i> 1998;79:334–341.	Confirmed (journal details; no DOI found)
19	Oran et al. Water temperature and inter-trial interval in the Morris water maze in aged female Wistar rats. <i>Turk J Geriatr</i> 2024.	Partial (abstract-level; title approximated)
20	Xiong, Semyanov, Tang. Commentary on Iram et al. <i>Signal Transduct Target Ther</i> 2022. doi:10.1038/s41392-022-01092-x	Confirmed by search summary

Removed because they could not be found: Wang Y et al., *Neurotherapeutics* 2026 (intranasal FGF17 in stroke); Gardner RS et al., *Behav Brain Res* 2026;496:115816 (only a 2024 bioRxiv preprint exists); Andero R, Choi DC, Ressler KJ, *Prog Mol Biol Transl Sci* 2014;122:1–31 (the closest real paper is Andero and Ressler, *Genes Brain Behav* 2012); the "7 months in monkeys" toxicity statement; the reagent-cost and budget figures.

Not found at all: a primary source for free-hand P7–P10 rat ICV injection and its mortality; juvenile (P21–P35) Alzet ICV protocols; age-stratified rat CSF volume (adult values range 150–300 μ l); a rat hippocampal TrkB.FL/T1 age series; a single source stating the myelination peak; a published effect-size anchor for aged-rat MWM treatment studies.